

PHYS-E0510 – Advanced Computational Methods in Physics D

Session 1: Classical Magnetism and Mean-Field Theory

Companion notebook: `classical_magnetism.ipynb` **Prerequisites:** none (second quantization is introduced from scratch)

Contents and goals

In this session we build the minimal machinery we need to talk about interacting electrons, second quantization, and we use it to answer a very concrete question: when do repulsive electron–electron interactions spontaneously break a symmetry of the Hamiltonian? We will look at two symmetry-broken phases, classical magnetism and charge density waves, and we will treat both of them entirely at the mean-field level. By the end of the session we will see a case where mean-field theory demonstrably gives us the wrong ground state, the Hubbard dimer, and that failure is precisely what motivates the exact-diagonalization and tensor-network methods we develop in the rest of the course.

Learning outcomes

By the end of this session, you should be able to:

- Write single-particle and many-body Hamiltonians in second-quantized form, and explain why the interaction term makes the many-body problem exponentially hard.
- Set up and run a mean-field self-consistency loop for the Hubbard and CDW models, and interpret the resulting exchange splitting, gap, and order parameter.
- Derive the superexchange coupling $J \sim t^2/U$ from degenerate perturbation theory and use it to predict collinear versus frustrated classical spin order.
- Identify when mean-field theory fails qualitatively, and explain why, using the entangled singlet ground state of the Hubbard dimer as the counterexample.

1 Single-particle versus many-body Hamiltonians

Let us start by distinguishing two broad classes of electronic Hamiltonians. The first class is what we call single-particle Hamiltonians: they are bilinear in the field operators,

$$\mathcal{H}_0 = \sum_{ij} t_{ij} c_i^\dagger c_j, \quad (1)$$

where t_{ij} is a hopping amplitude that we can think of as the probability amplitude for an electron in orbital j to jump to orbital i . Because $\mathcal{H}_0$ is just a matrix acting on a single-particle Hilbert space, every electron here moves essentially independently of the others, and finding the spectrum is nothing more than diagonalizing an $L \times L$ matrix for L orbitals. This is already enough to describe a wide variety of materials: insulators, semiconductors, ordinary metals.

The case we actually care about most, though, is the many-body case. Here we again have the

bilinear term, but we also add a term that is quartic in the fermionic operators,

$$\mathcal{H} = \mathcal{H}_0 + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j^\dagger c_k c_l, \quad (2)$$

and V_{ijkl} tells us the strength of this interaction. This is the term that is responsible for essentially all of the exotic phenomena we care about in quantum materials: magnetism, charge density waves, superconductivity, and every strongly correlated phase we will meet later in the course. What makes second quantization so useful is that it lets us write both the single-particle and the many-body piece of Eq. (2) in one common language.

2 Second quantization

So let us set up that language. We define the fermionic annihilation and creation operators c_i , $c_i^\dagger$ for orbital i , and a vacuum state $|\Omega\rangle$ that has no electrons at all, $c_i |\Omega\rangle = 0$ for every i . You can think of the vacuum as a vector and of the operators as matrices that map one vector to another; a single-particle basis state is simply $c_i^\dagger |\Omega\rangle$, and the matrix elements of $\mathcal{H}_0$ in this basis are exactly the hopping amplitudes t_{ij} we started with.

How do we enforce, mathematically, that no site can hold more than one electron of a given species? We impose the canonical anticommutation relations,

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \quad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \quad (3)$$

This single algebraic statement does two things at once: swapping two fermions picks up a minus sign, which is exactly antisymmetry of the wavefunction, and acting with $c_i^\dagger$ twice on any state gives zero, which is exactly the Pauli exclusion principle. Throughout the course we will meet three flavors of local degree of freedom built this way:

- **Spinful fermions:** each site has four possible states, $|0\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|\uparrow\downarrow\rangle$. This is the realistic case, the one you would use for something like a hydrogen atom.
- **Spinless fermions:** each site has only two states, $|0\rangle$ and $|1\rangle$. Formally, a spinful fermion is really nothing more than two decoupled spinless fermions, one per spin flavor.
- **Spin- $\frac{1}{2}$ operators:** again two states, $|\uparrow\rangle$ and $|\downarrow\rangle$, but now the operators are *bosonic*: they commute on different sites instead of anticommuting. As we will see in Section 6, a local spin emerges naturally from a spinful fermion once we project out double occupancy and the empty state. This is the origin of what is called spin-charge separation.

✕ Computational companion

Open `classical_magnetism.ipynb`, section “Bandstructure of a tight binding chain” and “Finite size chains.” Diagonalize $\mathcal{H}_0$ for a chain of L sites and check for yourself that the number of eigenvalues grows linearly in L , while their spacing shrinks: this is the discrete precursor of a continuous band. Then, in “Density of states of a finite system as a function of the system size,” plot the density of states, $\rho(\omega) = -\frac{1}{\pi} \text{Im Tr} [\omega - \mathcal{H}_0 + i0^+]^{-1}$, and think about how it connects to what a tunneling-spectroscopy or ARPES experiment actually measures.

3 Worked example: the tight-binding dimer

Before we add any interactions, it is worth solving the smallest possible chain by hand, since we will keep coming back to it. Take $L = 2$ sites with hopping t , and let us include spin from the start, so each site can hold a spin-up and a spin-down electron independently:

$$\mathcal{H}_0 = -t \sum_{\sigma} \left(c_{0\sigma}^{\dagger} c_{1\sigma} + c_{1\sigma}^{\dagger} c_{0\sigma} \right). \quad (4)$$

Spin never enters the hopping, so $\mathcal{H}_0$ splits into two identical 2×2 spinless problems, one per spin flavor:

$$\mathcal{H}_0^{(\sigma)} = -t \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \implies \varepsilon_{\pm} = \mp t, \quad |\pm\rangle = \frac{1}{\sqrt{2}} (|0\rangle \pm |1\rangle). \quad (5)$$

We call $|+\rangle$ the **bonding** orbital, at energy $-t$, and $|-\rangle$ the **antibonding** orbital, at energy $+t$. Now restore spin: each of these two orbital levels can separately hold a spin-up and a spin-down electron, so we actually have $2 \times 2 = 4$ single-particle eigenstates in total, even though there are only *two* distinct energies, $\pm t$, each doubly degenerate. This is the direct answer to a question you may have already asked yourself while looking at the notebook: why four eigenstates for two sites, and why are some of them degenerate? The degeneracy is counting spin, not orbital content, and it follows from the fact that $\mathcal{H}_0$ commutes with total spin: nothing in $\mathcal{H}_0$ distinguishes $\uparrow$ from $\downarrow$.

If we fill in electrons one at a time starting from $|\Omega\rangle$, we generate the full many-body basis for two spinful sites, exactly as we set up in the previous section. At half filling with two electrons, for instance, there are $\binom{4}{2} = 6$ states in total: four of them are singly occupied ($\uparrow\downarrow$, $\downarrow\uparrow$, $\uparrow\uparrow$, $\downarrow\downarrow$, one electron per site) and two of them are doubly occupied (both electrons on the same site). Keep this counting in mind, since it is exactly what we will use in Section 6 to build the strong-coupling effective spin model.

✕ Computational companion

Section “The spectra of a dimer.” Diagonalize the two-site, spinful tight-binding Hamiltonian and check that you recover the bonding/antibonding energies $\pm t$ and their spin degeneracy directly from the numerical spectrum.

4 Two microscopic interactions, two broken symmetries

Equation (2) is completely general, so in practice we have to pick a concrete form for V_{ijkl} and ask what symmetry-broken state it favors once we go to the thermodynamic limit, $L \rightarrow \infty$. We will focus on two of the simplest and most physically motivated choices.

The Hubbard model (spinful fermions). Suppose we only keep the interaction between opposite spins sitting on the *same* site:

$$\mathcal{H}_{\text{Hub}} = - \sum_{ij,\sigma} t_{ij} c_{i\sigma}^{\dagger} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad n_{i\sigma} \equiv c_{i\sigma}^{\dagger} c_{i\sigma}. \quad (6)$$

Physically, this local repulsion is what an on-site electrostatic (Coulomb) interaction reduces to once screening is included, and it is the minimal Hamiltonian we need for **magnetism**. Magnetic

order breaks time-reversal and spin-rotational symmetry: the exact Hamiltonian is invariant under any spin rotation, but its ground state is not, and if you apply time reversal to the ground state you get back a distinct state with the opposite magnetization.

The spinless nearest-neighbor model. Now drop the spin degree of freedom altogether and instead keep a density–density repulsion between neighboring sites:

$$\mathcal{H}_{\text{CDW}} = -t \sum_i \left(c_i^\dagger c_{i+1} + \text{h.c.} \right) + V \sum_i n_i n_{i+1}, \quad n_i \equiv c_i^\dagger c_i, \quad (7)$$

which is what you would get from a longer-range electrostatic repulsion once you truncate it to nearest neighbors. This is the minimal Hamiltonian for a **charge density wave (CDW)**: here the broken symmetry is the lattice’s *translational* symmetry, and charge piles up on one sublattice while it is depleted on the other.

4.1 Where do U and V come from physically?

We will treat U and V as given parameters for the rest of the course, but it is worth pausing on where they actually come from, because that also explains why both repulsive and attractive effective interactions are physically reasonable.

- **Direct electrostatics.** Screened Coulomb repulsion between electrons is always present, and it is always repulsive. Its on-site piece is what gives us U ; its longer-range piece, truncated to nearest neighbors, is what gives us V .
- **Quasiparticle-mediated interactions.** If we integrate out some other bosonic degree of freedom that couples to the electron density or spin, lattice vibrations (phonons), collective spin fluctuations (magnons), or collective charge fluctuations (plasmons, excitons), we generate an *effective* quartic interaction of exactly the same form as Eq. (2), even though there was no bare electron–electron term to begin with.

The interaction we actually see is the sum of all these contributions, and depending on which mechanism wins it can come out net repulsive or net *attractive*. In this course we work with repulsive $U, V > 0$, which is the regime relevant for magnetism and CDW order; attractive interactions instead favor Cooper pairing and superconductivity, a case we will not pursue here but which uses exactly the same formal machinery.

5 Mean-field theory

Solving Eq. (6) exactly means working in a Hilbert space of dimension 4^L (or 2^L per spin species), which is exponential in the system size. So the question we want to ask is not how to solve it exactly, but how to solve it approximately: is there some single-particle Hamiltonian $\mathcal{H}_{\text{MF}}$ whose ground state is close to the true ground state of $\mathcal{H}$? Finding that Hamiltonian is the whole art of mean-field theory.

5.1 Decoupling the interaction

The simplest way to do this is to write each quartic term as a product of two bilinears and replace one of the bilinears by its expectation value, dropping the fluctuation term that is left over (this is exact only at $U = 0$, but we will see it can still be a very good approximation away from that limit):

$$n_{i\uparrow}n_{i\downarrow} \longrightarrow \langle n_{i\uparrow} \rangle n_{i\downarrow} + n_{i\uparrow} \langle n_{i\downarrow} \rangle - \langle n_{i\uparrow} \rangle \langle n_{i\downarrow} \rangle. \quad (8)$$

If we define the local magnetization as $m_i \equiv \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle$, the Hubbard interaction turns into an effective, spin-dependent on-site potential:

$$\mathcal{H}_{\text{MF}} = - \sum_{ij,\sigma} t_{ij} c_{i\sigma}^\dagger c_{j\sigma} - \frac{U}{2} \sum_i m_i \sigma_z + \text{const.} \quad (9)$$

where $\sigma_z = \pm 1$ for $\sigma = \uparrow, \downarrow$, so the on-site term reads $-\frac{U}{2}m_i\sigma$ site by site. So spin-up electrons now see a shifted on-site energy $-Um_i/2$, and spin-down electrons see $+Um_i/2$. This splitting, which is generated entirely by the interaction, is what we call the **exchange splitting**. More generally, if we also keep expectation values like $\langle c_{i\uparrow}^\dagger c_{i\downarrow} \rangle$, we let $\mathcal{H}_{\text{MF}}$ describe non-collinear magnetization, with components $m_i^x \equiv 2 \text{Re} \langle c_{i\uparrow}^\dagger c_{i\downarrow} \rangle$, $m_i^y \equiv 2 \text{Im} \langle c_{i\uparrow}^\dagger c_{i\downarrow} \rangle$, and $m_i^z \equiv m_i$, so in principle mean-field theory can capture ferromagnets, antiferromagnets, spin spirals, or even skyrmion textures.

The self-consistency loop in practice. Concretely, here is what we actually do:

1. Guess an initial value for the local order parameters, e.g. $m_i^{(0)}$ for magnetism or $\langle n_i \rangle^{(0)}$ for a CDW.
2. Build $\mathcal{H}_{\text{MF}}$ from Eq. (9) using this guess, and diagonalize it.
3. Fill the lowest single-particle orbitals up to the target electron number, and recompute m_i (or $\langle n_i \rangle$) from the newly occupied orbitals.
4. Compare the new order parameters to the previous ones. If they have not changed, within some tolerance, we are done: this is the self-consistent solution. Otherwise, mix the old and new guesses and go back to step 2.

Because this is a nonlinear fixed-point problem, it can converge to different fixed points depending on where you start it, and that is exactly the initial-guess dependence we come back to below.

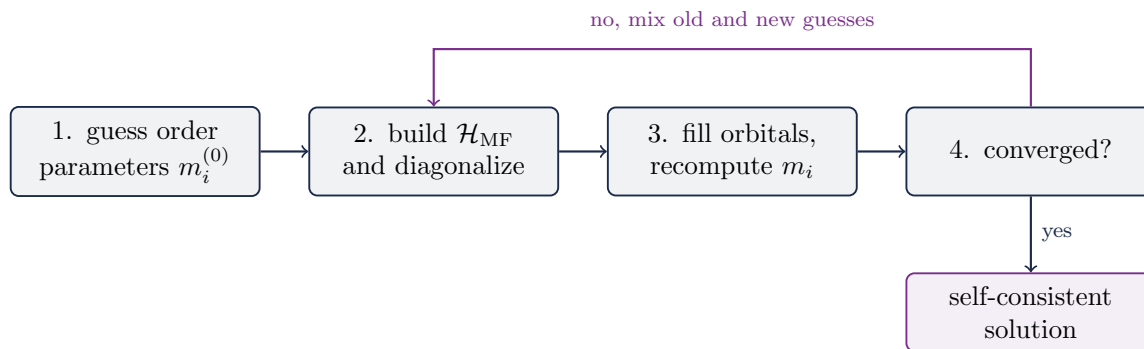


Figure 1: The self-consistency loop, which is what every mean-field calculation in this session actually runs. Steps 2 and 3 are the two halves that alternate: an order parameter fixes a single-particle Hamiltonian through Eq. (9), and diagonalizing that Hamiltonian and filling its lowest orbitals up to the target electron number fixes a new order parameter. Convergence means the two agree to within a tolerance, and until they do we mix the old and new guesses and go around again.

5.2 Mean-field theory as a variational method

There is an equivalent, and arguably more illuminating, way to think about all of this. Suppose we restrict the many-body ground state to be a single Slater determinant, $|\Psi\rangle = c_{a_1}^\dagger c_{a_2}^\dagger \cdots |\Omega\rangle$, and we minimize $\langle \mathcal{H} \rangle$ over the choice of single-particle orbitals $\{a_n\}$. Since $\mathcal{H}_{\text{MF}}$ depends on $\langle n_{i\sigma} \rangle$, and $\langle n_{i\sigma} \rangle$ itself depends on the ground state of $\mathcal{H}_{\text{MF}}$, minimizing the energy this way is precisely the self-consistency loop we just described: guess $\{m_i\}$, diagonalize $\mathcal{H}_{\text{MF}}$, recompute $\{m_i\}$ from the filled orbitals, and repeat until it converges. So mean-field theory is really the best approximation we can build out of *product states*, and that restriction to product states is exactly what will get us into trouble in Section 9.

✕ Computational companion

Section “Magnetic order driven by interactions in a chain.” Run the self-consistency loop for a chain at fixed filling and look at the converged spin-up/spin-down bands: you should see one cosine band per spin, split by the exchange field, with only the spin-down band occupied at the filling used in the notebook.

5.3 Gap opening at half filling

Let us now force half filling and double the unit cell, so we have two sites with alternating majority spin. At $U = 0$ this just folds the metallic cosine band onto itself. Once $U > 0$, the self-consistent solution opens a gap between the occupied and empty bands, and that gap grows with U , eventually becoming linear, $\Delta \sim U$, once $U \gg t$. If you look at the single-particle spectral function instead of the eigenvalues, you see exactly the same story: a continuous metallic density of states at $U = 0$ develops a clean gap that widens with U , which is precisely what a spectroscopy experiment would pick up.

5.4 Multiple self-consistent solutions and their energetics

Because the self-consistency loop is a *nonlinear* fixed-point problem, the solution it converges to depends on where we start it. A ferromagnetic initial guess converges to a metallic, ferromagnetic solution; an antiferromagnetic initial guess converges to an insulating, antiferromagnetic solution. Both of these are legitimate stationary points of the mean-field energy, but only one of them is the true, lowest-energy mean-field ground state. If we take a half-filled chain with only nearest-neighbor hopping, it turns out the antiferromagnetic solution is always lower in energy, and the gap between the two scales as

$$E_{\text{FM}} - E_{\text{AFM}} \sim \frac{1}{U} \xrightarrow{U \rightarrow \infty} 0. \quad (10)$$

So the two solutions only become degenerate in the strict atomic limit $U \rightarrow \infty$; at any finite U they are physically distinct states, one metallic and one insulating.

✕ Computational companion

Section “The interacting Hubbard dimer.” Compare a ferromagnetic-seeded and an antiferromagnetic-seeded self-consistent run on the dimer of Section 3, now with $U > 0$ switched on, and check numerically which solution has the lower variational energy and how the gap between them scales with U . It is also worth revisiting “Gap opening in the Hubbard model” with a ferromagnetic initialization, to see this initial-guess dependence show up directly in the spectrum.

6 The strong-coupling limit: from electrons to spins

As $U/t \rightarrow \infty$, something nice happens to the local moment $\langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle$: it saturates, in magnitude, to 1 on every site. Double occupancy and empty sites become energetically forbidden, and every site ends up carrying exactly one localized electron. This is the regime where the physics is no longer best described by itinerant fermions, but by *spins*, and it is worth understanding in detail why.

6.1 The Hubbard dimer and its low-energy manifold

Take our two-site dimer again, with hopping t , interaction U , and two electrons total (so we are at half filling). We already counted a six-dimensional many-body basis for this problem in Section 3. For $U \gg t$, this basis splits cleanly into two groups:

- a four-dimensional **ground-state manifold** of singly-occupied configurations ($\uparrow\downarrow, \downarrow\uparrow, \uparrow\uparrow, \downarrow\downarrow$ across the two sites), sitting at energy ~ 0 ;
- a **virtual (high-energy) manifold** of doubly-occupied configurations, sitting at energy $\sim U$.

Look closely at the ground-state manifold: it is exactly the Hilbert space of two spin- $\frac{1}{2}$ degrees of freedom. In other words, our strongly interacting electronic problem has quietly turned into a spin problem.

6.2 Superexchange: $J = 4t^2/U$

The only term that connects the two manifolds is the hopping t , and symmetry alone (there is no spin-rotation breaking anywhere in $\mathcal{H}_{\text{Hub}}$) already tells us that the low-energy effective Hamiltonian has to be proportional to $\mathbf{S}_1 \cdot \mathbf{S}_2$ at leading order in spin operators. To actually pin down the coefficient, we treat $\mathcal{H}_0 = U \sum_i n_{i\uparrow} n_{i\downarrow}$ as the unperturbed Hamiltonian, with the singly- and doubly-occupied configurations of Section 3 as its eigenstates, and the hopping $V = -t \sum_{\sigma} (c_{0\sigma}^\dagger c_{1\sigma} + \text{h.c.})$ as the perturbation. The parallel configurations $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ have no doubly-occupied virtual state to hop into at all, since Pauli exclusion blocks it, so they are exact eigenstates already, unshifted at any order: this is our triplet, sitting at $E_{\text{triplet}} = 0$.

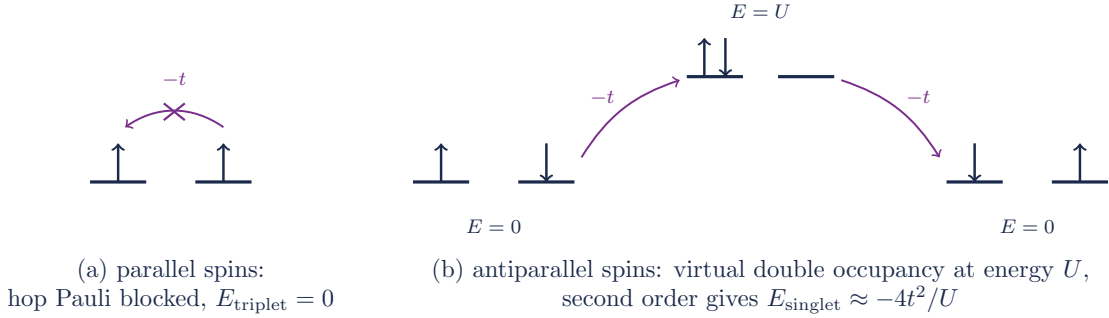


Figure 2: The two spin channels of the half-filled Hubbard dimer at strong coupling, which is where the $J = 4t^2/U$ of Eq. (13) comes from. (a) With parallel spins there is no doubly-occupied state available to hop into, Pauli exclusion blocks the process outright, and the triplet stays exactly at $E_{\text{triplet}} = 0$ at every order. (b) With antiparallel spins either electron can hop, passing through a virtual doubly-occupied configuration at energy U and landing on the exchanged configuration. Each hop carries a matrix element $-t$ and the virtual state is paid for by an energy denominator $-U$, so second-order perturbation theory lowers the antiparallel channel by an amount of order t^2/U while leaving the parallel one untouched.

The antiparallel configurations are more subtle, since *either* electron can hop, so each one of $|\uparrow\downarrow\rangle = c_{0\uparrow}^\dagger c_{1\downarrow}^\dagger |\Omega\rangle$ and $|\downarrow\uparrow\rangle = c_{0\downarrow}^\dagger c_{1\uparrow}^\dagger |\Omega\rangle$ connects through V to *both* virtual doubly-occupied states, $|2, 0\rangle = c_{0\uparrow}^\dagger c_{0\downarrow}^\dagger |\Omega\rangle$ and $|0, 2\rangle = c_{1\uparrow}^\dagger c_{1\downarrow}^\dagger |\Omega\rangle$, each sitting at energy U above the low-energy manifold, with matrix elements $\langle 2, 0 | V | \uparrow\downarrow \rangle = -t$ and $\langle 0, 2 | V | \uparrow\downarrow \rangle = -t$ (and, by the site-exchange symmetry of the problem, $\langle 2, 0 | V | \downarrow\uparrow \rangle = +t$, $\langle 0, 2 | V | \downarrow\uparrow \rangle = +t$, the sign flipping because of fermionic anticommutation once the two electrons are reordered onto the same site). It is only the total spin, not the sublattice, that is conserved here, so the clean way to organize the calculation is to work directly in the singlet/triplet basis rather than $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$. Only the singlet combination $|s\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ has any overlap with the doubly-occupied states at all (the triplet combination's matrix elements to $|2, 0\rangle$ and $|0, 2\rangle$ cancel by the sign flip above), and by the same site-exchange symmetry only the symmetric combination $|d\rangle = \frac{1}{\sqrt{2}}(|2, 0\rangle + |0, 2\rangle)$ couples back to it, since the antisymmetric combination $\frac{1}{\sqrt{2}}(|2, 0\rangle - |0, 2\rangle)$ is odd under the mirror that $|s\rangle$ is even under. Explicitly,

$$\langle d | V | s \rangle = \frac{1}{\sqrt{2}} [\langle 2, 0 | V | \uparrow\downarrow \rangle - \langle 2, 0 | V | \downarrow\uparrow \rangle + \langle 0, 2 | V | \uparrow\downarrow \rangle - \langle 0, 2 | V | \downarrow\uparrow \rangle] \frac{1}{\sqrt{2}} = \frac{1}{2} [-t - t - t - t] = -2t, \quad (11)$$

so the singlet sector reduces to an exact 2×2 problem, $|s\rangle$ at energy 0 coupled to $|d\rangle$ at energy U by the matrix element $-2t$, with the orthogonal antisymmetric doubly-occupied combination

decoupled entirely, remaining an exact eigenstate at energy U . Standard non-degenerate second-order (Schrieffer–Wolff, or equivalently Löwdin) perturbation theory on this 2×2 block gives

$$E_{\text{singlet}} \approx 0 + \frac{|\langle d|V|s \rangle|^2}{0 - U} = -\frac{4t^2}{U}, \quad (12)$$

which one can check is exactly the $U \gg t$ expansion of the closed-form eigenvalue of the full 2×2 matrix, $E_- = \frac{U}{2} - \sqrt{(U/2)^2 + 4t^2} \rightarrow -4t^2/U$; a full numerical exact diagonalization of the two-site, two-electron Hubbard model confirms both this asymptotic value and the exact triplet energy $E_{\text{triplet}} = 0$ at every U/t , not just at strong coupling. Compare this to the exact spectrum of $J \mathbf{S}_1 \cdot \mathbf{S}_2$ itself: writing $\mathbf{S}_{\text{tot}} = \mathbf{S}_1 + \mathbf{S}_2$ gives $\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2} [S_{\text{tot}}(S_{\text{tot}} + 1) - \frac{3}{4} - \frac{3}{4}]$, which evaluates to $-\frac{3}{4}$ for the singlet ($S_{\text{tot}} = 0$) and $+\frac{1}{4}$ for the triplet ($S_{\text{tot}} = 1$), i.e. $E_{\text{singlet}} = -\frac{3}{4}J$ and $E_{\text{triplet}} = \frac{1}{4}J$, a gap of exactly J between them. Matching this gap to the one we just computed fixes

$$J = E_{\text{triplet}} - E_{\text{singlet}} = \frac{4t^2}{U} \quad \implies \quad \mathcal{H}_{\text{eff}} = J \mathbf{S}_1 \cdot \mathbf{S}_2, \quad (13)$$

the **antiferromagnetic superexchange** coupling. Intuitively: virtual hopping into the doubly-occupied states, which is Pauli-blocked for parallel spins, lowers the energy of the singlet combination of antiparallel configurations, and that is why the effective low-energy interaction ends up favoring antialignment even though the underlying Hubbard repulsion has no explicit spin dependence at all. The same argument goes through unchanged on an arbitrary lattice with hoppings t_{ij} : the half-filled, strongly interacting Hubbard model on *any* lattice maps onto a spin- $\frac{1}{2}$ **Heisenberg model**,

$$\mathcal{H}_{\text{Heis}} = \sum_{ij} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j, \quad J_{ij} \sim \frac{t_{ij}^2}{U}. \quad (14)$$

with the same numerical prefactor as Eq. (13), $J_{ij} = 4t_{ij}^2/U$, for each bond treated in isolation this way. If instead we have more than one electron per site, as in multi-orbital atoms, we map onto higher-spin Heisenberg models instead: spin-1 for two electrons per site, spin- $\frac{3}{2}$ for three, and so on. Additional weakly interacting orbitals can even flip the *sign* of J_{ij} , giving us ferromagnetic rather than antiferromagnetic coupling, which is worth keeping in mind since real materials are not always antiferromagnets.

✕ Computational companion

Section “The strongly interacting limit of the Hubbard model.” Compute $E_{\text{FM}} - E_{\text{AFM}}$ as a function of U and check that it follows the $1/U$ scaling we derived above. Then push to smaller U and see where the perturbative assumption $t \ll U$ behind Eq. (13) starts to break down.

7 Classical magnetic order and frustration

If we now treat the spins in $\mathcal{H}_{\text{Heis}}$ as classical vectors, an approximation that is in spirit very close to mean-field theory itself, finding the ground state becomes a straightforward energy minimization.

Bipartite lattices. On the square and honeycomb lattices, we can always split the sites into two sublattices A, B such that every bond connects A to B . For antiferromagnetic $J > 0$, the classical ground state is then simply $\mathbf{S}_A = -\mathbf{S}_B$: collinear Néel order.

Frustration. Things get more interesting on lattices that are *not* bipartite, like the triangular and kagome lattices, which are built out of corner- or edge-sharing triangles. There, no assignment of two sublattices can satisfy every antiferromagnetic bond at once. The classical ground state instead becomes **non-collinear**: neighboring spins rotate by 120° as you go around each triangle, producing a spiral magnetic order with a multi-site magnetic unit cell.

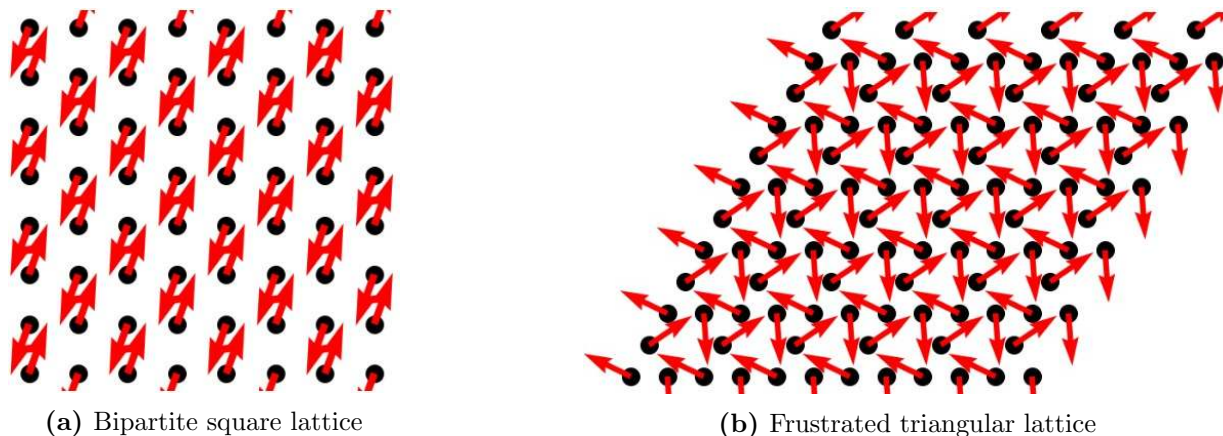


Figure 3: Classical ground states of the Heisenberg model of Eq. (14) with antiferromagnetic $J > 0$, obtained by treating every spin as a classical vector. (a) On a bipartite lattice every bond connects the two sublattices, so $\mathbf{S}_A = -\mathbf{S}_B$ satisfies all of them simultaneously and the order is collinear Néel. (b) On the triangular lattice the bonds sit on corner-sharing triangles and no two-sublattice assignment can satisfy all of them, so the minimum-energy configuration is instead the non-collinear 120° spiral, with a magnetic unit cell spanning several sites.

✕ Computational companion

Section “Non-collinear magnetism in the Hubbard model from competing interactions” and “Non-collinear magnetic order from geometric frustration.” Add a second-neighbor hopping to the mean-field chain and see that it, too, can stabilize non-collinear order: competing exchange paths play the same frustrating role as triangular geometry does. Then move to the triangular/kagome supercells and read off the 120° spiral directly from the self-consistent magnetization pattern.

Anisotropic exchange. Real materials often break the $SU(2)$ spin-rotation symmetry of $\mathcal{H}_{\text{Heis}}$ through spin–orbit coupling, adding bilinear-in-spin terms that are *not* symmetric under a common rotation of all spins. There are three that come up often enough to be worth naming:

- **Antisymmetric (Dzyaloshinskii–Moriya) exchange**, $\mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j)$, which shows up when mirror symmetry is broken and favors spins sitting at 90° to each other, i.e. non-collinear order even without any geometric frustration.
- **Ising anisotropy**, coupling only $S_i^z S_j^z$, which arises from strong spin–orbit coupling and single-ion anisotropy and pushes the spins toward classical behavior, confined to a single axis or plane.
- **Kitaev exchange**, a *bond-dependent* Ising-like coupling where x -bonds couple $S^x S^x$, y -bonds couple $S^y S^y$, and z -bonds couple $S^z S^z$. No single spin configuration can satisfy all three bond types at once on, say, the honeycomb lattice, which is exactly why this term is intrinsically frustrating and a key ingredient behind quantum spin liquid models.

None of these three terms are covered in the companion notebook. We flag them here as slides-only material that becomes relevant again in more advanced treatments of quantum magnetism.

8 Charge density waves as a mean-field problem

The very same machinery applies to $\mathcal{H}_{\text{CDW}}$ from Eq. (7): we decouple the density–density term via $\langle n_i \rangle n_{i+1} + n_i \langle n_{i+1} \rangle - \langle n_i \rangle \langle n_{i+1} \rangle$ and iterate to self-consistency, exactly as before. At $V = 0$ the chain is a featureless, half-filled metal with $n_i = \frac{1}{2}$ on every site. As we turn on V , we get exactly the same qualitative spectral evolution as in the Hubbard case: a gap opens and grows $\sim V$ at large V . What changes is the order parameter, which is now the **charge imbalance**,

$$\delta n \equiv n_{\text{even}} - n_{\text{odd}}, \quad (15)$$

and it grows from 0 at $V = 0$ toward $\delta n \rightarrow 1$ at large V , meaning the sites end up alternating fully filled and fully empty. The broken symmetry here is translational rather than time-reversal, but notice how closely the mechanism, and even the shape of the spectral evolution, mirrors the magnetic case.

✕ Computational companion

Sections “Gap opening in a spinless interacting model” and “Symmetry breaking in the spinless fermionic chain.” Track the gap and the charge imbalance δn as a function of V , then seed the self-consistency loop with an explicit external CDW field and see how it selects which sublattice orders.

9 Where mean-field theory breaks down

So far mean-field theory has worked well every time, and that is really because it is a very good technique precisely when the true ground state *does* spontaneously break a symmetry. But nothing guarantees this in advance, and the Hubbard dimer at half filling gives us a clean counterexample.

The exact Hamiltonian $\mathcal{H}_{\text{Hub}}$ for two sites is fully spin-rotation and time-reversal symmetric. Its mean-field decoupling, however, is only self-consistent with a *symmetry-breaking* ansatz once U passes a threshold: below threshold, $\mathcal{H}_{\text{MF}}$ stays non-magnetic, and above it, the converged solution has finite, eventually saturating local moments of opposite sign on the two sites, exactly the regime we studied in Section 6. The mean-field gap tracks this transition too, staying constant below threshold and then growing linearly above it.

Now compare this to the *exact* many-body ground state, and the story looks quite different. In the strongly interacting limit, Section 6 already told us the dimer maps onto $\mathcal{H}_{\text{eff}} = J \mathbf{S}_1 \cdot \mathbf{S}_2$, and we can just diagonalize that by hand: its exact ground state is the singlet

$$|\Psi_{\text{singlet}}\rangle = \frac{1}{\sqrt{2}} \left(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right). \quad (16)$$

This state has $\langle S_1^z \rangle = \langle S_2^z \rangle = 0$. It simply does not break time-reversal symmetry, in direct conflict with what the symmetry-broken mean-field solution told us to expect. The reason is structural, not a numerical accident: $|\Psi_{\text{singlet}}\rangle$ is an entangled superposition, and it cannot be

written as any single Slater determinant, no matter what choice of orbitals or spin quantization axis we try. Mean-field theory, built by construction out of product states (Section 6), can simply never reach it.

Why this matters. The Hubbard dimer is the smallest Hamiltonian you can write down that shows this failure mode, but the failure itself is completely generic: whenever the true many-body ground state is genuinely entangled and not adiabatically connected to a product state, mean-field theory gets the physics qualitatively wrong, the wrong symmetry, the wrong order, sometimes an entirely wrong phase. This is exactly the agenda for the rest of the course: *exact diagonalization* (Sessions 2–3) solves small many-body Hamiltonians with no product-state restriction at all, and *tensor network* methods (Sessions 4–5) let us recover the exact answer at system sizes where exact diagonalization is no longer feasible.

	Mean-field theory	Exact diagonalization	Tensor networks (MPS)
Ansatz	product state (Slater det.)	unrestricted, full Hilbert space	bounded-entanglement state (MPS)
Cost vs. size L	polynomial	exponential, $\sim 2^L$	polynomial in L and bond dimension
Captures entanglement?	no	yes, exactly	yes, up to the bond dimension
Typical reach	any L	$L \lesssim 20\text{--}30$	$L \sim 40\text{--}100+$ (1D)
Fails when...	ground state is not adiabatically connected to a product state (this session)	never, in principle (only limited by L)	entanglement exceeds what the bond dimension can store

Table 1: The three methods used in this course, previewed side by side: their ansatz, cost scaling with system size L , whether they capture entanglement, typical reach, and failure mode.

Think of Table 1 as a preview rather than a derivation: we will develop and use each of these methods on its own terms starting next session.

Summary

- Second quantization lets us write single-particle and many-body ($\mathcal{O}(4)$ -fermion) Hamiltonians in one language; the interaction term is what drives all correlated physics.
- Mean-field theory decouples the interaction into an effective single-particle problem that we solve self-consistently; equivalently, it is a variational search over Slater determinants.
- The Hubbard model gives us magnetism (broken time-reversal/spin symmetry); the spinless nearest-neighbor model gives us charge density waves (broken translational symmetry), and both come from essentially the same mean-field mechanism.
- In the strongly interacting limit, second-order perturbation theory maps the Hubbard model onto a Heisenberg spin model with $J \sim t^2/U$, the superexchange coupling. Classically this model shows collinear order on bipartite lattices and non-collinear/frustrated order otherwise.

- Mean-field theory can fail qualitatively whenever the true ground state is entangled beyond any product state, exactly as we saw for the Hubbard dimer's singlet ground state. That failure is what motivates exact diagonalization and tensor networks, which we take up next session.